VENDOR RECONCILIATION — Technical Blueprint

SAP CAP & UI5 Accounts Payable Analytics Platform

PROJECT BLUEPRINT
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Executive Summary: The Vendor Reconciliation Platform is an Accounts Payable (AP) analytics and matching engine built on SAP CAP (Node.js) and SAP UI5. It automatically compares incoming vendor invoices against corresponding Purchase Orders, mathematically detecting price and quantity variances at the line-item level, and surfaces the results through an Executive Dashboard and a detailed Object Page.

SECTION 1 — CURRENT IMPLEMENTATION

Accounts Payable teams often struggle to manually verify line-item quantities and gross amounts across hundreds of vendor invoices. **Our Build:** We have constructed an automated backend Matching Engine that links Invoices to POs, calculates discrepancies, and flags exceptions. This data is visualized via an SAP UI5 Executive Dashboard, providing real-time KPIs and the ability to drill down into a specific invoice's line items for variance breakdown.

SECTION 2 — CORE ARCHITECTURE

Data Flow & Evaluation: The system ingests real, live ERP data directly from the SAP S21 system (via the S21 Connector module) and runs it through the matching algorithm before exposing it to the UI.

1. S21 Connector
Node.js module (`s21-connector.js`) configured to fetch POs and Invoices from SAP.

2. Matching Engine
(`matching-engine.js`) iterates through Line Items, comparing File Qty vs S/4 Qty.

3. CAP OData Services
Exposes the resulting `MatchResults` and aggregated KPIs via OData V4 endpoints.

4. UI5 Frontend
Consumes the OData services to render the Dashboard and Object Page views.

HIGH-LEVEL ARCHITECTURE DIAGRAM



SECTION 3 — IMPLEMENTED CAPABILITIES

◆ **Executive Dashboard (Cockpit)**
A high-level Fiori page displaying 4 critical KPIs (Processed, Exception Rate, Critical Variances, Match Rate) powered by a live backend KPI handler.

◆ **Client-Side PDF Export**
A functional "Export PDF" button on the Dashboard utilizing `html2canvas` and `jspdf` to take a pixel-perfect snapshot of the analytics UI.

◆ **Variance Identification**
The backend engine dynamically identifies if a discrepancy is a "Qty Mismatch" or "Price Mismatch" and calculates the exact `amountDelta`.

◆ **Detailed Object Page**
A deep-dive view comparing the Vendor File header values against the SAP S/4 records side-by-side, complete with a Line Item variance table.

◆ **AI Intelligence & Scoring**
The workbench provides intelligent **AI Suggestions** and computes a holistic **AI Score** for each matched invoice, assisting the AP team in determining the safest course of action.

◆ **Actionable Invoice States**
Empowers AP to set invoice statuses directly in the workbench:
- **Approved:** Ready to proceed for payment.
- **Blocked:** Payment stopped due to critical blockages.
- **Parked:** Held for unresolved issues. (Not posted to SAP, but provides AP a centralized actionable view).

SECTION 4 — FRONTEND UI COMPONENTS

The user interface is built entirely on **SAP UI5** (XML Views and JavaScript Controllers), ensuring a modern enterprise look and feel.

Component	Implementation Details
Cockpit.view.xml	Uses `sap.m.Page` and `sap.m.Toolbar`. Renders KPI cards mathematically computed by the backend, along with a recent Match Results table.
ObjectPage.view.xml	Utilizes `sap.uxap.ObjectPageLayout`. Stripped of OCR elements per requirements, focusing strictly on ERP data comparison (Company Code, Vendor Code, Tax Amounts).
style.css	Custom CSS file defining the Light Theme (Slate-50 background, Indigo typography, and rounded 12px card elevations).
Formatter (formatter.js)	Client-side logic handling currency formatting and converting S/4HANA date strings (YYYY-MM-DD) into readable UI formats.

SECTION 5 — BACKEND DATA MODEL (SCHEMA.CDS)

The CAP Application uses a relational database schema designed to hold ingested ERP documents and the results of the matching engine.

Entity Name	Purpose & Attributes
VendorInvoiceHeaders	Stores the high-level invoice data (Invoice Number, Gross Amount, `invoiceDate`, Tax Amount).
VendorDocLine	Stores line-level details of the invoice (File Qty, File Price, Material Code).
PurchaseOrders	Stores the ERP source-of-truth (PO Number, Target Qty, Expected Price).
MatchResults	The output ledger of the matching engine. Contains `varianceAmount`, `status` (Matched/Warning), and arrays of Line Comparisons.

SECTION 6 — DATA BINDING & ROUTING

UI5 Routing Engine: The application utilizes the standard UI5 `sap.m.routing.Router`. The primary route (`main`) points to the Executive Dashboard. Clicking an Invoice Number in the Dashboard's table triggers a deep-link route (`#/invoice/{resultId}`) which loads the `ObjectPage.view.xml`.

OData V4 Orchestration: In `ObjectPage.controller.js`, the view executes dynamic OData V4 `$expand` queries to fetch `MatchResults`, expanding into the related `vendorDocLine`, `invoiceItems`, and `purchaseOrders` to construct the complete context for the UI.

SECTION 7 — TECHNOLOGY STACK

FRONTEND UI
SAP UI5 Framework
XML Views, JS Controllers
Custom CSS Customizations

BACKEND CORE
Node.js & Express
SAP Cloud Application Programming (CAP)

DATABASE
SQLite (Local persistence)
Core Data Services (CDS) Definitions

SECTION 8 — BUSINESS KEY SELLING POINTS

1. Eradication of Manual Effort
Completely automates the grueling line-by-line comparison of invoices against purchase orders and goods receipts, saving thousands of AP labor hours monthly.

2. Zero Tolerance for Overbilling
The mathematical matching engine catches identical invoice duplicates and micro-variances in pricing and quantity before they are paid, directly preventing revenue leakage.

3. AI-Driven Decision Support
Instead of raw data dumps, the platform provides actionable AI Suggestions and a holistic AI Score, empowering AP clerks to process complex discrepancies with confidence.

4. Single Pane of Glass
Unifies Parked, Blocked, and Approved invoices into one centralized dashboard. Financial controllers gain immediate, real-time visibility into the entire AP pipeline without needing native SAP GUI access.